# MSG No. 213.63 $P4_132$ [ Type I, cubic ]

Table 1: Wyckoff site: `4a`, site symmetry: `.32`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$ | `[1,12,14,16,17,18]` |
| 2 | $\left[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}\right]$ | `[2,7,9,15,19,24]` |
| 3 | $\left[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}\right]$ | `[3,4,10,11,20,21]` |
| 4 | $\left[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}\right]$ | `[5,6,8,13,22,23]` |

Table 2: Wyckoff site: `4b`, site symmetry: `.32`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$ | `[1,12,14,16,17,18]` |
| 2 | $\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$ | `[2,7,9,15,19,24]` |
| 3 | $\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$ | `[3,4,10,11,20,21]` |
| 4 | $\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$ | `[5,6,8,13,22,23]` |

Table 3: Wyckoff site: `8c`, site symmetry: `.3.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, x]$ | `[1,17,18]` |
| 2 | $\left[x+\frac{1}{4}, \frac{1}{4}-x, x+\frac{3}{4}\right]$ | `[2,7,15]` |
| 3 | $\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-x\right]$ | `[3,4,11]` |
| 4 | $\left[\frac{1}{4}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$ | `[5,6,13]` |
| 5 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$ | `[8,22,23]` |
| 6 | $\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[9,19,24]` |
| 7 | $\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$ | `[10,20,21]` |
| 8 | $\left[\frac{3}{4}-x, \frac{3}{4}-x, \frac{3}{4}-x\right]$ | `[12,14,16]` |

Table 4: Wyckoff site: `12d`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{8}, y, y+\frac{1}{4}\right]$ | `[1,13]` |
| 2 | $\left[\frac{3}{8}, -y, y+\frac{3}{4}\right]$ | `[2,10]` |
| 3 | $\left[\frac{7}{8}, y+\frac{1}{2}, \frac{1}{4}-y\right]$ | `[3,9]` |
| 4 | $\left[y, y+\frac{1}{4}, \frac{1}{8}\right]$ | `[4,18]` |
| 5 | $\left[-y, y+\frac{3}{4}, \frac{3}{8}\right]$ | `[5,19]` |
| 6 | $\left[\frac{1}{4}-y, \frac{7}{8}, y+\frac{1}{2}\right]$ | `[6,20]` |
| 7 | $\left[y+\frac{1}{4}, \frac{1}{8}, y\right]$ | `[7,17]` |
| 8 | $\left[\frac{5}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$ | `[8,14]` |
| 9 | $\left[y+\frac{3}{4}, \frac{3}{8}, -y\right]$ | `[11,22]` |
| 10 | $\left[\frac{3}{4}-y, \frac{5}{8}, \frac{1}{2}-y\right]$ | `[12,24]` |
| 11 | $\left[y+\frac{1}{2}, \frac{1}{4}-y, \frac{7}{8}\right]$ | `[15,23]` |
| 12 | $\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{5}{8}\right]$ | `[16,21]` |

Table 5: Wyckoff site: 24e, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[x+\frac{1}{4},\ \frac{1}{4}-z,\ y+\frac{3}{4}\right]$ | [2] |
| 3 | $\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{4}-y\right]$ | [3] |
| 4 | $\left[z+\frac{3}{4},\ y+\frac{1}{4},\ \frac{1}{4}-x\right]$ | [4] |
| 5 | $\left[\frac{1}{4}-z,\ y+\frac{3}{4},\ x+\frac{1}{4}\right]$ | [5] |
| 6 | $\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$ | [6] |
| 7 | $\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{3}{4}\right]$ | [7] |
| 8 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$ | [8] |
| 9 | $\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [9] |
| 10 | $\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$ | [10] |
| 11 | $\left[y+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$ | [11] |
| 12 | $\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$ | [12] |
| 13 | $\left[\frac{1}{4}-x,\ z+\frac{3}{4},\ y+\frac{1}{4}\right]$ | [13] |
| 14 | $\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ \frac{3}{4}-y\right]$ | [14] |
| 15 | $\left[z+\frac{1}{4},\ \frac{1}{4}-y,\ x+\frac{3}{4}\right]$ | [15] |
| 16 | $\left[\frac{3}{4}-z,\ \frac{3}{4}-y,\ \frac{3}{4}-x\right]$ | [16] |
| 17 | $[z,\ x,\ y]$ | [17] |
| 18 | $[y,\ z,\ x]$ | [18] |
| 19 | $\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$ | [19] |
| 20 | $\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$ | [20] |
| 21 | $\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$ | [21] |
| 22 | $\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$ | [22] |
| 23 | $\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$ | [23] |
| 24 | $\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$ | [24] |